

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		16643	
Total Threats Mitigated	62	Total Packets Analyzed		3198029126	
High Priority	19	Malicious Packets		3187833589	
Low Priority	43	True Negatives Passed		12328	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		17	
ML Metrics	Value	SYN Flood		24	
False Positives	0	UDP Flood		11	
FP Rate	0.00%	Uncertain		10	
IF Precision	100.00%				
IF Recall	95.24%	Action		Count	
IF F1-Score	97.56%	Quarantined		10	
IF Accuracy	99.12%	Time Ban		10	
		Blackhole		2	
		Released		5	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.24%	Share of actual attacks successfully flagged
F1-Score	97.56%	Balanced measure combining Precision and Recall
Accuracy	99.12%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.76%	Actual attacks that went undetected
True Positive Rate (TPR)	95.24%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 2623	FN 131
Actual: Normal	FP 0	TN 12197

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	1516	0	0
Act: ICMP	0	89	0
Act: UDP	0	0	1018

2c. Hybrid Cascaded

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.24%	Share of actual attacks successfully flagged
F1-Score	97.56%	Balanced measure combining Precision and Recall
Accuracy	99.12%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.76%	Actual attacks that went undetected
True Positive Rate (TPR)	95.24%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 2623	FN 131
Actual: Normal	FP 0	TN 12197

2d. Response Latency

Metric	Value	Description
Detection Time	372.9 ms	Average time taken to detect an attack
Mitigation Response Time	5.4 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	4.16%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	6.00%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	5.41%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.17	8	5	2026-09-25 14:35:57	2026-09-25 14:42:17
10.0.0.23	7	5	2026-09-25 14:35:58	2026-09-25 14:41:57
10.0.0.16	6	5	2026-09-25 14:35:59	2026-09-25 14:41:54
10.0.0.20	6	5	2026-09-25 14:35:59	2026-09-25 14:41:54
10.0.0.22	6	5	2026-09-25 14:36:00	2026-09-25 14:42:03
10.0.0.24	6	5	2026-09-25 14:35:59	2026-09-25 14:42:02
10.0.0.25	6	5	2026-09-25 14:35:59	2026-09-25 14:42:05
10.0.0.19	5	5	2026-09-25 14:35:57	2026-09-25 14:40:08

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.18	3	2	2026-09-25 14:35:57	2026-09-25 14:39:55
10.0.0.21	2	2	2026-09-25 14:39:52	2026-09-25 14:42:01

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 14:35:57	10.0.0.17	DDoS	SYN Flood	92.2%	Critical	Quarantined
2026-09-25 14:35:57	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Quarantined
2026-09-25 14:35:57	10.0.0.18	DDoS	SYN Flood	93.3%	Critical	Quarantined
2026-09-25 14:35:58	10.0.0.23	DDoS	ICMP Flood	91.2%	High	Time Ban (30s)
2026-09-25 14:35:58	10.0.0.23	DDoS	ICMP Flood	91.2%	High	Time Ban
2026-09-25 14:35:59	10.0.0.20	DDoS	UDP Flood	76.3%	High	Time Ban (30s)
2026-09-25 14:35:59	10.0.0.20	DDoS	UDP Flood	76.3%	High	Time Ban
2026-09-25 14:35:59	10.0.0.25	DDoS	ICMP Flood	90.1%	High	Time Ban (30s)
2026-09-25 14:35:59	10.0.0.25	DDoS	ICMP Flood	90.1%	High	Time Ban
2026-09-25 14:35:59	10.0.0.24	DDoS	ICMP Flood	90.0%	High	Time Ban (30s)
2026-09-25 14:35:59	10.0.0.24	DDoS	ICMP Flood	90.0%	High	Time Ban
2026-09-25 14:36:00	10.0.0.22	DDoS	UDP Flood	86.1%	Critical	Quarantined
2026-09-25 14:36:08	10.0.0.17	DDoS	SYN Flood	92.2%	Critical	Time Ban (30s)
2026-09-25 14:36:08	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Time Ban (30s)
2026-09-25 14:36:08	10.0.0.18	DDoS	SYN Flood	93.3%	High	Released
2026-09-25 14:36:08	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-25 14:36:09	10.0.0.16	DDoS	SYN Flood	89.3%	Critical	Time Ban (30s)
2026-09-25 14:36:11	10.0.0.22	DDoS	UDP Flood	86.1%	Critical	Time Ban (30s)
2026-09-25 14:37:08	10.0.0.17	DDoS	SYN Flood	92.2%	Critical	Time Ban (1m)
2026-09-25 14:37:08	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-25 14:37:09	10.0.0.16	DDoS	SYN Flood	89.3%	Critical	Time Ban (1m)
2026-09-25 14:37:11	10.0.0.22	DDoS	UDP Flood	86.1%	Critical	Time Ban (1m)
2026-09-25 14:37:29	10.0.0.23	DDoS	ICMP Flood	91.2%	High	Time Ban (1m)
2026-09-25 14:37:29	10.0.0.20	DDoS	UDP Flood	76.3%	High	Time Ban (1m)
2026-09-25 14:37:30	10.0.0.25	DDoS	ICMP Flood	90.1%	High	Time Ban (1m)
2026-09-25 14:37:30	10.0.0.24	DDoS	ICMP Flood	90.0%	High	Time Ban (1m)
2026-09-25 14:38:08	10.0.0.17	DDoS	SYN Flood	92.2%	Critical	Time Ban (2m)
2026-09-25 14:38:08	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)
2026-09-25 14:38:09	10.0.0.16	DDoS	SYN Flood	89.3%	Critical	Time Ban (2m)
2026-09-25 14:38:11	10.0.0.22	DDoS	UDP Flood	86.1%	Critical	Time Ban (2m)
2026-09-25 14:38:29	10.0.0.23	DDoS	ICMP Flood	91.2%	High	Time Ban (2m)
2026-09-25 14:38:30	10.0.0.25	DDoS	ICMP Flood	90.1%	High	Time Ban (2m)
2026-09-25 14:38:30	10.0.0.24	DDoS	ICMP Flood	90.0%	High	Time Ban (2m)
2026-09-25 14:39:16	10.0.0.17	DDoS	SYN Flood	92.2%	Critical	Time Ban
2026-09-25 14:39:16	10.0.0.16	DDoS	SYN Flood	89.3%	Critical	Time Ban
2026-09-25 14:39:25	10.0.0.18	DDoS	SYN Flood	93.3%	Critical	Time Ban (30s)
2026-09-25 14:39:25	10.0.0.18	DDoS	SYN Flood	93.3%	Critical	Time Ban

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 14:39:29	10.0.0.20	DDoS	UDP Flood	76.3%	High	Time Ban (2m)
2026-09-25 14:39:45	10.0.0.22	DDoS	UDP Flood	86.1%	Critical	Time Ban
2026-09-25 14:39:52	10.0.0.21	DDoS	UDP Flood	85.6%	High	Time Ban (30s)
2026-09-25 14:39:52	10.0.0.21	DDoS	UDP Flood	85.6%	High	Time Ban
2026-09-25 14:39:55	10.0.0.18	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-09-25 14:40:08	10.0.0.17	DDoS	SYN Flood	92.2%	Critical	Time Ban (5m)
2026-09-25 14:40:08	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Time Ban (5m)
2026-09-25 14:40:09	10.0.0.16	DDoS	SYN Flood	89.3%	Critical	Time Ban (5m)
2026-09-25 14:40:29	10.0.0.23	DDoS	ICMP Flood	96.1%	Critical	Time Ban (5m)
2026-09-25 14:40:30	10.0.0.25	DDoS	ICMP Flood	96.1%	Critical	Time Ban (5m)
2026-09-25 14:40:30	10.0.0.24	DDoS	ICMP Flood	96.1%	Critical	Time Ban (5m)
2026-09-25 14:41:44	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:41:44	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:41:45	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:41:50	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:41:51	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:41:54	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (10m)
2026-09-25 14:41:55	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Blackhole
2026-09-25 14:41:55	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:41:55	10.0.0.23	DDoS	ICMP Flood	88.3%	Critical	Time Ban (10m)
2026-09-25 14:41:57	10.0.0.23	DDoS	ICMP Flood	89.6%	Critical	Blackhole
2026-09-25 14:42:01	10.0.0.21	DDoS	Uncertain	-	Low	Released
2026-09-25 14:42:02	10.0.0.24	DDoS	Uncertain	-	Low	Released
2026-09-25 14:42:03	10.0.0.22	DDoS	Uncertain	-	Low	Released
2026-09-25 14:42:05	10.0.0.25	DDoS	Uncertain	-	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6259	94.2%	Critical	Phase 3	1	1s	2026-09-25 14:42:17	Escalated to Blackhole
10.0.0.25	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:42:05	Attack Stopped
10.0.0.22	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 14:42:03	Attack Stopped
10.0.0.24	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 14:42:02	Attack Stopped
10.0.0.21	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 14:42:01	Attack Stopped
10.0.0.23	ICMP Flood	0.6431	89.6%	Critical	Phase 3	2	12s	2026-09-25 14:41:57	Escalated to Blackhole

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.23	ICMP Flood	0.6431	88.3%	Critical	Phase 2	1	10s	2026-09-25 14:41:55	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6407	90.9%	Critical	Phase 3	2	11s	2026-09-25 14:41:55	Escalated to Blackhole
10.0.0.20	UDP Flood	0.6441	79.7%	Critical	Phase 2	1	10s	2026-09-25 14:41:54	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6416	90.9%	Critical	Phase 2	1	10s	2026-09-25 14:41:54	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6347	92.7%	Critical	Phase 2	1	10s	2026-09-25 14:41:54	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6401	96.1%	Critical	Phase 2	5	4m 31s	2026-09-25 14:40:30	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6401	96.1%	Critical	Phase 2	5	4m 31s	2026-09-25 14:40:30	Escalated to Time Ban
10.0.0.23	ICMP Flood	0.6413	96.1%	Critical	Phase 2	5	4m 31s	2026-09-25 14:40:29	Escalated to Time Ban
10.0.0.21	UDP Flood	0.6267	85.6%	High	Phase 2	2	31s	2026-09-25 14:40:23	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6446	86.1%	Critical	Phase 2	5	4m 11s	2026-09-25 14:40:11	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6523	89.3%	Critical	Phase 2	5	4m 10s	2026-09-25 14:40:09	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6163	93.1%	Critical	Phase 2	5	4m 11s	2026-09-25 14:40:08	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6476	92.2%	Critical	Phase 2	5	4m 11s	2026-09-25 14:40:08	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6123	93.3%	Critical	Phase 2	2	41s	2026-09-25 14:39:55	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6452	76.3%	High	Phase 2	5	3m 30s	2026-09-25 14:39:29	Escalated to Time Ban
10.0.0.18	SYN Flood	0.6487	93.3%	Critical	Phase 2	1	11s	2026-09-25 14:39:25	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6466	90.0%	High	Phase 2	5	2m 31s	2026-09-25 14:38:30	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6462	90.1%	High	Phase 2	5	2m 31s	2026-09-25 14:38:30	Escalated to Time Ban
10.0.0.20	UDP Flood	0.6452	76.3%	High	Phase 2	5	2m 30s	2026-09-25 14:38:29	Escalated to Time Ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.23	ICMP Flood	0.6443	91.2%	High	Phase 2	5	2m 31s	2026-09-25 14:38:29	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6466	86.1%	Critical	Phase 2	4	2m 11s	2026-09-25 14:38:11	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6547	89.3%	Critical	Phase 2	4	2m 10s	2026-09-25 14:38:09	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6190	93.1%	Critical	Phase 2	4	2m 11s	2026-09-25 14:38:08	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6372	92.2%	Critical	Phase 2	4	2m 11s	2026-09-25 14:38:08	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6466	90.0%	High	Phase 2	4	1m 31s	2026-09-25 14:37:30	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6462	90.1%	High	Phase 2	4	1m 31s	2026-09-25 14:37:30	Escalated to Time Ban
10.0.0.20	UDP Flood	0.6452	76.3%	High	Phase 2	4	1m 30s	2026-09-25 14:37:29	Escalated to Time Ban
10.0.0.23	ICMP Flood	0.6443	91.2%	High	Phase 2	4	1m 31s	2026-09-25 14:37:29	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6466	86.1%	Critical	Phase 2	3	1m 11s	2026-09-25 14:37:11	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6547	89.3%	Critical	Phase 2	3	1m 10s	2026-09-25 14:37:09	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6239	93.1%	Critical	Phase 2	3	1m 11s	2026-09-25 14:37:08	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6372	92.2%	Critical	Phase 2	3	1m 11s	2026-09-25 14:37:08	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6466	90.0%	High	Phase 2	3	1m 1s	2026-09-25 14:37:00	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6462	90.1%	High	Phase 2	3	1m 1s	2026-09-25 14:37:00	Escalated to Time Ban
10.0.0.20	UDP Flood	0.6452	76.3%	High	Phase 2	3	1m 0s	2026-09-25 14:36:59	Escalated to Time Ban
10.0.0.23	ICMP Flood	0.6443	91.2%	High	Phase 2	3	1m 1s	2026-09-25 14:36:59	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6466	86.1%	Critical	Phase 2	2	41s	2026-09-25 14:36:41	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6547	89.3%	Critical	Phase 2	2	40s	2026-09-25 14:36:39	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6309	93.1%	Critical	Phase 2	2	41s	2026-09-25 14:36:38	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6372	92.2%	Critical	Phase 2	2	41s	2026-09-25 14:36:38	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6466	90.0%	High	Phase 2	2	31s	2026-09-25 14:36:30	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6462	90.1%	High	Phase 2	2	31s	2026-09-25 14:36:30	Escalated to Time Ban
10.0.0.20	UDP Flood	0.6452	76.3%	High	Phase 2	2	30s	2026-09-25 14:36:29	Escalated to Time Ban
10.0.0.23	ICMP Flood	0.6443	91.2%	High	Phase 2	2	31s	2026-09-25 14:36:29	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6466	86.1%	Critical	Phase 2	1	11s	2026-09-25 14:36:11	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6547	89.3%	Critical	Phase 2	1	10s	2026-09-25 14:36:09	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6080	93.3%	High	Phase 1	1	11s	2026-09-25 14:36:08	Attack Stopped
10.0.0.19	SYN Flood	0.6371	93.1%	Critical	Phase 2	1	11s	2026-09-25 14:36:08	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6372	92.2%	Critical	Phase 2	1	11s	2026-09-25 14:36:08	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE